

BABU BANARASI DAS **UNIVERSITY**



SCHOOL OF COMPUTER APPLICATIONS

ASSIGNMENT ON DESCRIPTIVE ANALYSIS (BCADSN23201)

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DESCRIPTIVE ANALYTICS REPORT

OLA/UBER CAB SERVICE

KEYWORDS-

Fare Distribution, Cab Type Preference, Customer Rating, City-wise Demand, Distance-Fare Relationship, Ola/Uber, Cab Service Operations.

TIME FRAME-

The dataset represents a snapshot of 100 cab trips collected over a period of 2 months, providing a representative sample of short-term booking patterns, fare trends, and city-wise demand across the 10 major Indian cities covered in this study.

NOTE ON DATA-

The ride-hailing industry in India has grown rapidly, with platforms like Ola and Uber becoming a primary mode of urban transportation. **This report presents a descriptive analysis of a dataset representing 100 cab trips across 10 major Indian cities, including Mumbai, Delhi, Bangalore, Hyderabad, and Lucknow.**

The dataset captures key attributes such as fare amount, trip distance, cab type, customer rating, booking status, and payment method. Descriptive analytics helps us understand the basic characteristics of data by summarising and visualising patterns. **This report applies descriptive statistical techniques to extract meaningful insights about cab usage behaviour, fare distribution, and service quality.**

India has one of the largest and fastest-growing ride-hailing markets in the world. With millions of trips happening every day across major cities, **companies like Ola and Uber generate enormous amounts of data.** Analysing this data helps businesses

make better decisions related to pricing, driver allocation, and customer satisfaction.

How Does This Work at Real-Life Scale?

Processing millions of cab trips daily is an engineering and analytics challenge. Platforms like Ola and Uber handle crores of trips every day across hundreds of Indian cities. To make sense of this massive amount of data, a structured data pipeline is used. This pipeline collects, processes, stores, and visualises trip data in real time and on a daily basis. The report below reflects the output of such a pipeline using a sample dataset of 100 trips.

The following table explains each layer of the real-world data pipeline used by large cab service companies:

Layer:	Purpose:	Typical Tools:
➤ Ingestion	Every time a customer books a cab, the trip details like location, cab type, fare, and time are captured instantly in real time.	Google Forms / Mobile App
➤ Stream Processing	Raw trip events are rolled up into per-city, per-cab-type, and per-minute aggregates for quick analysis.	Microsoft Excel / Python
➤ Storage	Both raw trip data and pre-aggregated summary tables are stored for audits and fast querying.	Google Drive / MySQL
➤ Batch Aggression	Every night, daily rollups are created — total trips, average fares, ratings, and city-wise demand.	Python / Jupyter Notebook

➤ Serving/BI Layer	Final reports are created and refreshed on an hourly or daily basis, allowing business teams to track key metrics and monitor performance in real time.	MS Excel
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In our report, we have used a synthetic dataset of 100 trip records that mirrors the kind of aggregated output this pipeline would produce. The same analytical structure — fare distribution, cab type analysis, city-wise demand, ratings, and distance-fare relationship — applies directly to real-world Ola/Uber data at scale. Once real data is substituted, the same charts and findings would reflect actual business performance across millions of daily trips.

RESEARCH OBJECTIVES:

The primary aim of this report is to conduct a descriptive analysis of Ola/Uber cab service data in order to understand key patterns and trends in urban cab usage across Indian cities. By analyzing 100 trip records, this report seeks to answer five important research questions related to fare distribution, cab type preferences, customer satisfaction, city-wise demand, and the relationship between trip distance and fare amount.

These research questions have been carefully selected to cover all major aspects of cab service operations — from pricing and vehicle preferences to service quality and geographic demand. The findings from this analysis can provide valuable insights to cab service companies, helping them improve their operations, optimize pricing strategies, and enhance overall customer experience.

Q1. What is the average fare amount and how is it distributed across trips?

Q2. Which cab type is most frequently booked by customers?

Q3. How are customer ratings distributed and what is the overall service quality?

Q4. Which city has the highest number of trips?

Q5. Is there a relationship between trip distance and fare amount?

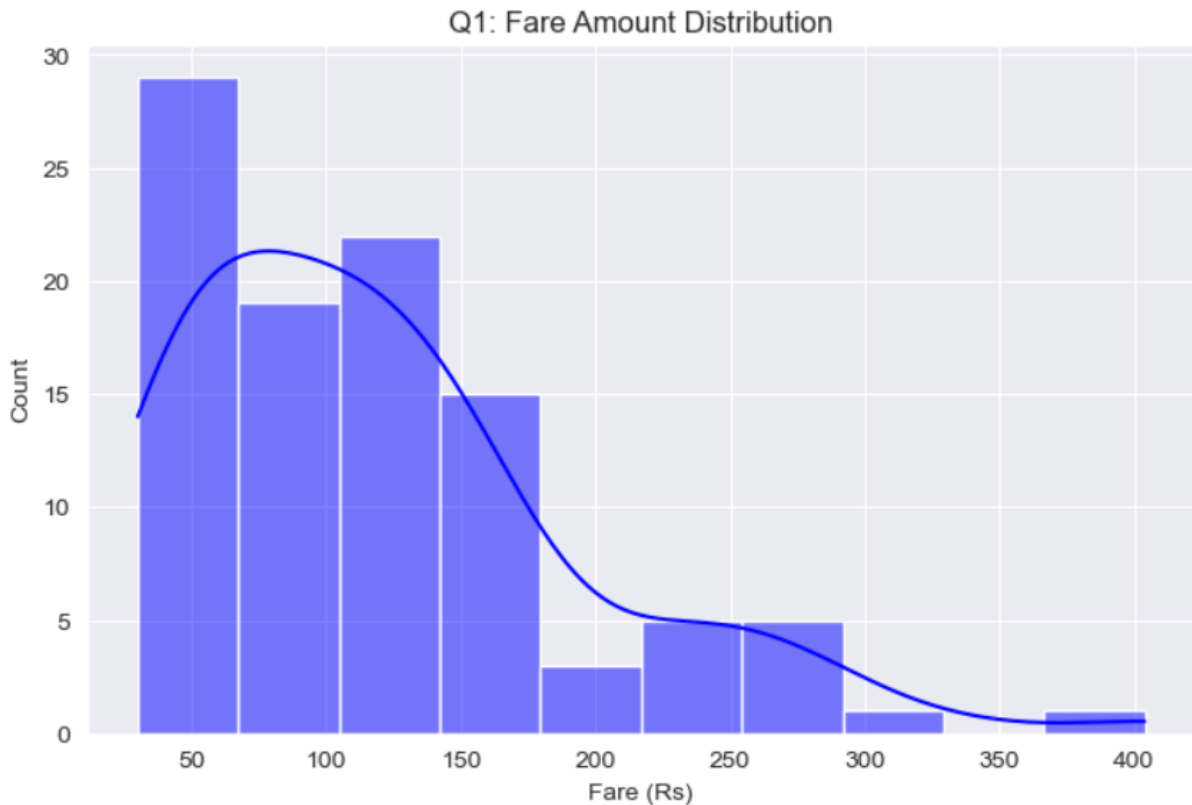
Fare Amount Distribution

The fare amount across **100 trips** was analysed using measures of central tendency and a histogram chart.

Average (Mean) Fare: Rs. 119.6, Median Fare: Rs. 105.5, Mode Fare: Rs. 30.

KPI:

- Average Fare: **Rs. 119.6**
- Median Fare: **Rs. 105.5**
- Min Fare: **Rs. 30** | Max Fare: **Rs. 404**



Finding (i): The fare distribution is right-skewed, meaning most trips have lower fares in the range of **Rs. 30–150**, while a few long-distance trips have significantly higher fares up to **Rs. 400**. The mean fare (**Rs. 119.6**) is higher than the median (**Rs. 105.5**), which confirms the right skew. This pattern is typical in cab services where short city trips dominate but occasional airport or outstation trips pull the average up.

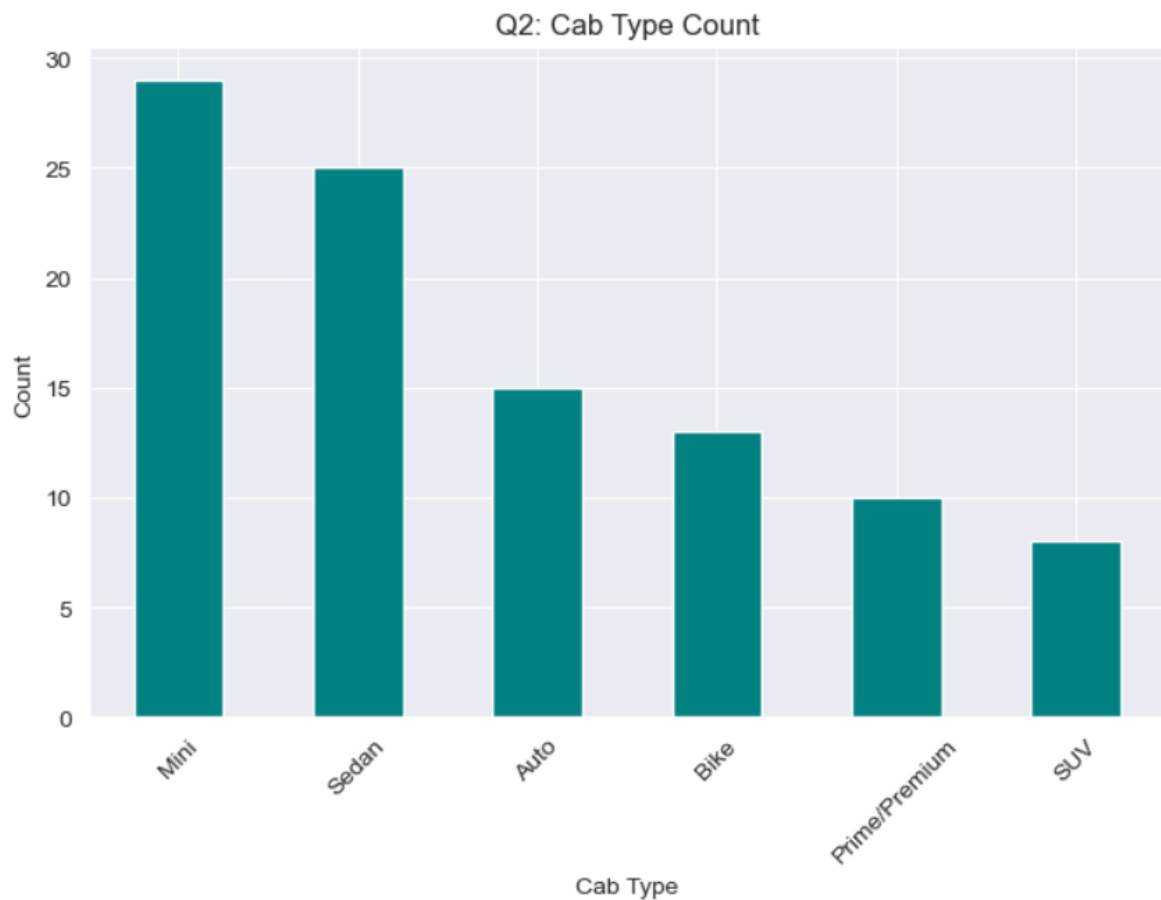
Most Frequently Booked Cab Type

The frequency of bookings by cab type was analysed using a bar chart.

📌 KPI:

- Most Booked: **Mini (29 trips)**
- Least Booked: **SUV (8 trips)**

- Total Cab Types: **6**



Finding (ii): Mini cabs are the most frequently booked with **29 trips**, followed by Sedan with **25 trips**. Auto and Bike come next. Prime/Premium and SUV are least booked as they are more expensive options. This reflects customer preference for affordable cab options.

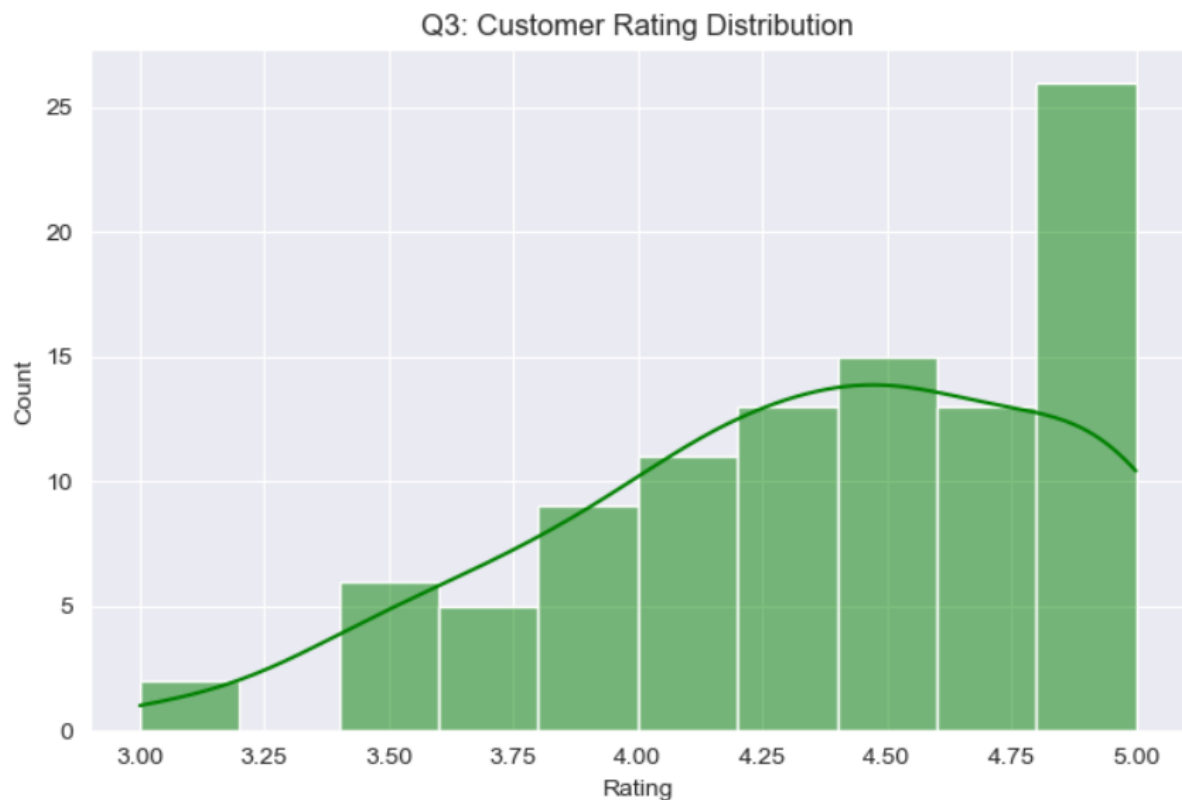
Customer Rating Distribution

Customer ratings were analysed to assess overall service quality across all trips.

📌 **KPI:**

- Average Rating: **4.34 / 5**

- Median Rating: **4.4 / 5**
- Highest Rating: **5.0** | Lowest Rating: **3**.



Finding (iii): The rating distribution shows that most customers gave high ratings between **4.0 and 5.0**. Very few trips received ratings below **3.5**, indicating overall high service quality. An average rating of **4.34** reflects strong customer satisfaction with Ola/Uber services.

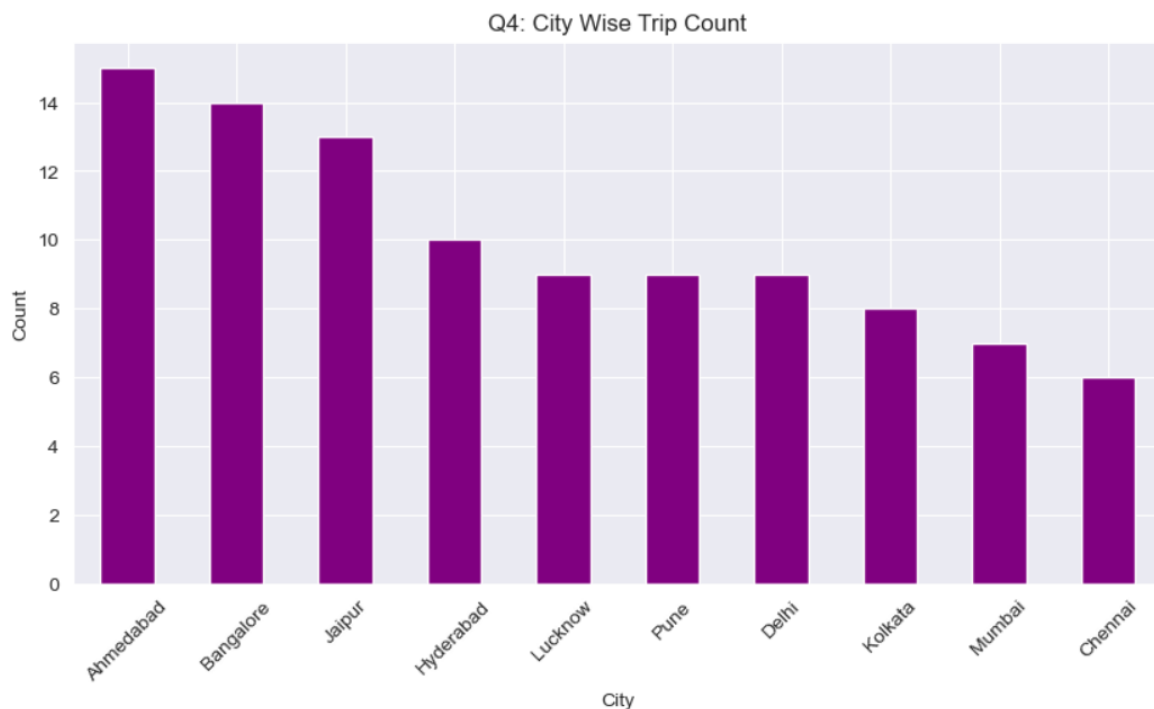
City-wise Trip Counting

The number of trips per city was analysed to identify demand concentration across Indian cities.

📌 KPI:

- Top City: **Ahmedabad (15 trips)**
- Lowest City: **Chennai (6 trips)**

- Total Cities **Covered: 10**



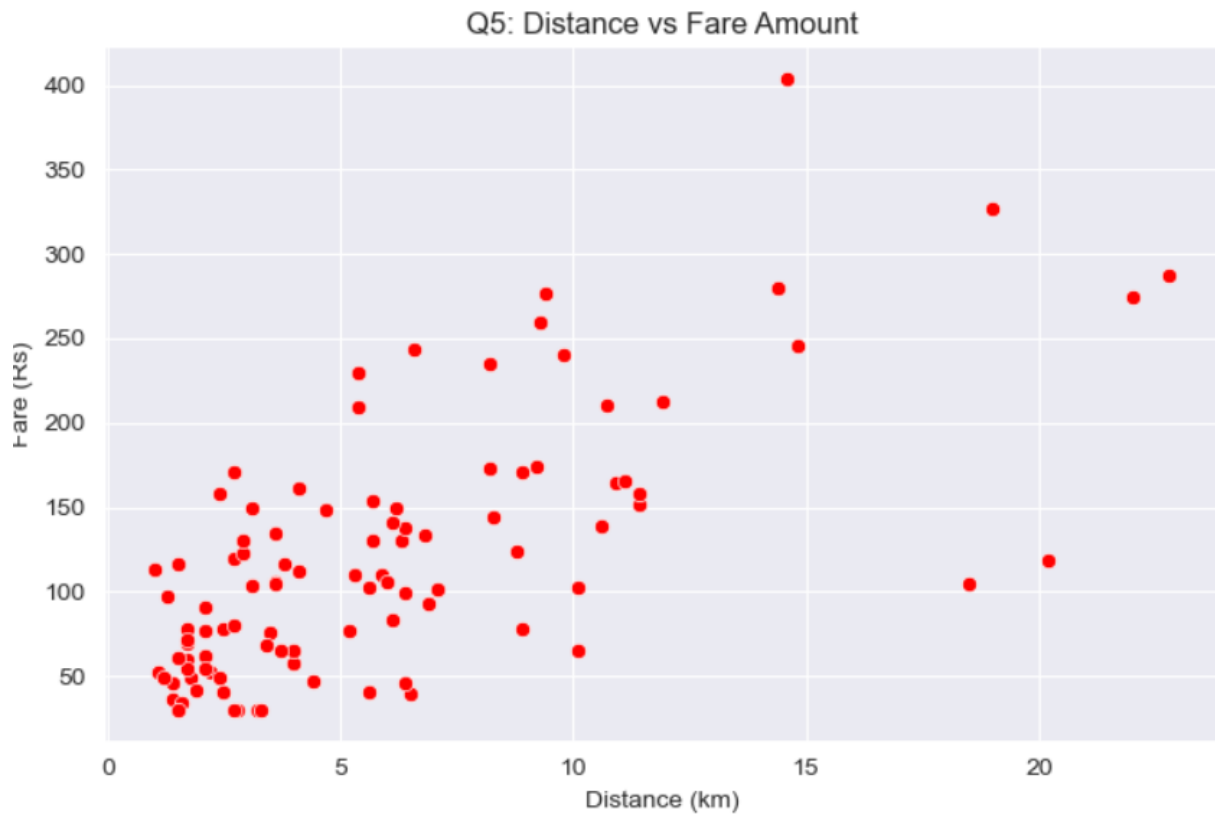
Finding (iv): Ahmedabad leads with **15 trips**, followed by Bangalore with 14 trips and Jaipur with **13 trips**. Metro cities like **Mumbai (7)** and **Chennai (6)** show lower trip counts in this dataset. This insight can help cab services plan driver deployment and pricing strategies city-wise.

Relationship Between Distance and Fare Amount

A scatter plot was used to examine whether a relationship exists between trip distance and fare amount.

📌 KPI:

- Average Distance: **5.97 km**
- Min Distance: **1 km** | Max Distance: **22.8 km**
- Positive correlation between distance and fare ☒



Finding (v): The scatter plot shows a positive relationship between distance and fare — as trip distance increases, fare amount also increases. However, there is some variation at shorter distances (**1–5 km**) where fares range from **Rs. 30 to Rs. 170** due to differences in cab type. Overall, distance is a strong factor in determining fare amount.

SUMMARY OF KEY METRICS

The following table summarizes the key findings from the descriptive analysis of 100 Ola/Uber cab trips across 10 major Indian cities. The analysis covers fare distribution, cab type preferences, customer ratings, city-wise demand, and the relationship between distance and fare amount.

Key-Metrics Table:

Metric:	Value:
➤ Total Trips Analysed	100
➤ Average Fare Amount	Rs. 119.6
➤ Median Fare Amount	Rs. 105.5
➤ Average Trip Distance	5.97 km
➤ Average Trip Distance	4.34 / 5
➤ Most Popular Cab Type	Mini (29 trips)
➤ City with Most Trips	Ahmedabad (15 trips)

➤ Cities Covered	10
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FUTURE SCOPE : From Predictive to Descriptive

This report covers descriptive analysis of Ola/Uber cab data using a synthetic dataset of 100 trips. While this analysis provides valuable insights into fare distribution, cab preferences, customer ratings, and city-wise demand, there is significant scope for further research and advanced analysis in this domain.

- **Predictive Analysis:** Machine learning models such as Linear Regression and Random Forest can be used to predict fare amount based on distance, cab type, city, and time of day. This would help cab companies set dynamic and optimised pricing.
- **Real-time Data Integration:** Live Ola/Uber trip data can be integrated using APIs and streaming tools like Apache Kafka. This would allow real-time monitoring of trips, fares, and driver availability across cities.
- **Customer Segmentation:** Customers can be grouped into segments based on trip purpose, payment method, frequency of travel, and preferred cab type using clustering algorithms like K-Means. This would help companies offer personalised promotions and services.
- **Sentiment Analysis:** Customer reviews, ratings, and feedback from app stores and social media can be analysed using Natural Language Processing (NLP) techniques to

understand satisfaction levels and identify areas of improvement.

- **Demand Forecasting:** Trip demand can be predicted city-wise, time-wise, and season-wise using time series forecasting models. This would help cab companies allocate drivers efficiently during peak hours and festive seasons
- In summary, this report serves as a starting point for a much deeper and more comprehensive analysis of cab service data. As data availability and analytical capabilities improve, the insights derived from such studies will play a crucial role in shaping the future of urban transportation in India.

REFERENCES-

The following table lists all the references, tools, and technologies used in the preparation of this report. This report has been prepared using Python programming language and its associated data analysis libraries for dataset generation and visualization. The dataset used in this report is a synthetic dataset created specifically for academic analysis purposes, designed to mirror real-world Ola/Uber cab service data. All charts, graphs, and statistical outputs presented in this report have been generated using Jupyter Notebook environment with the help of Pandas, NumPy, Matplotlib, and Seaborn libraries. The final report has been documented and formatted using Microsoft Word.

REFERENCE TABLE:

Sr no.	Reference	Details
➤ 1	Dataset	Synthetic Ola/Uber Trip Dataset (Generated for academic analysis)

➤ 2	Programming Lang.	Python 3
➤ 3	Tools Used	Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn
➤ 4	Report Format	Microsoft Word (.docx)

APPENDIX-

This appendix presents a representative sample of the raw dataset used for this analysis, illustrating the structure of the 100 trip records referenced throughout the report. The table below shows 10 sample trips across the key variables described in the Dataset Description section.

Sample Data (10 of 100 Trip Records):

**Trip_id City Cab_type Distance_km Fare_amount Trip_duration_min
Customer_rating Booking_status Payment_method Trip_purpose**

101. Ahmedabad Mini 3.2 65 14 4.5 Completed UPI Office
102. Mumbai Sedan 8.7 210 28 4.2 Completed Card Airport
103. Bangalore Auto 2.1 30 9 4.8 Completed Cash Shopping
104. Jaipur SUV 12.4 340 40 4.0 Completed Wallet Airport
105. Lucknow Mini 4.6 85 18 4.6 Completed UPI Office
106. Ahmedabad Bike 1.8 30 7 4.3 Completed Cash Shopping
107. Hyderabad Sedan 9.5 230 32 3.9 Cancelled Card Office
108. Chennai Mini 5.0 95 20 4.4 Completed UPI Office
109. Bangalore Auto 3.0 45 12 4.7 Completed Cash Shopping
110. Delhi SUV 14.2 400 45 4.1 No Show Wallet Airport

Note: The above records are illustrative samples drawn from the full 100-trip dataset and are consistent with the summary statistics presented in this report.

DATA ANALYSIS PROCESS / WORKFLOW

The analysis followed a simple workflow to identify patterns and trends in the Ola/Uber dataset of 100 synthetic trip records.

1. Data Collection

Used a synthetic dataset of 100 trips containing fare, distance, cab type, ratings, booking status, and payment method.

2. Data Processing

The data was organized and prepared for analysis.

3. Descriptive Analysis

Mean, median, mode, minimum, and maximum values were used to summarize the data.

4. Data Visualisation

Histograms, bar charts, and scatter plots were used to visualise the data.

5. Analysis and Interpretation

The results were interpreted to identify patterns in fares, cab preferences, ratings, city-wise demand, and distance-fare relationships.

6. Findings

Key findings were summarised to provide insights into cab service usage and customer behaviour.

Thank You!

For your Time and Attention

THIS REPORT WAS PREPARED AS PART OF THE DESCRIPTIVE ANALYTICS SUBJECT REQUIREMENT. THE ENTIRE ANALYSIS — FROM DATASET CREATION TO VISUALISATION AND DOCUMENTATION — WAS PERFORMED INDEPENDENTLY USING PYTHON, JUPYTER NOTEBOOK, AND MICROSOFT WORD.

WE HOPE THIS REPORT PROVIDES MEANINGFUL INSIGHTS INTO OLA/UBER CAB SERVICE DATA AND DEMONSTRATES THE PRACTICAL APPLICATION OF DESCRIPTIVE ANALYTICS TECHNIQUES.

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